

Derivation of the bare electron mass from $\{\epsilon_0, \mu_0, \hbar\}$ via Cosserat-functional minimization: $m_e^{\text{bare}} = 507.997 \text{ keV}$

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Abstract

The electron mass $m_e c^2 = 510.998950 \pm 0.000015 \text{ keV}$ (CODATA 2018) is an empirical input in the Standard Model; no derivation from more fundamental constants exists.

In the present work m_e is computed numerically from three vacuum constants $\{\epsilon_0, \mu_0, \hbar\}$ by minimizing the Cosserat functional $E[n, u]$ of a director field $n: \mathbb{R}^3 \rightarrow S^2$ over a three-parameter Hopf ansatz with topological charge $Q_H = 1$. All constants of the functional are fixed by structural identities from the preceding work [1]:

- lattice scale $l_0^4 = \hbar / (2\pi Z_0)$, with $Z_0 = \sqrt{(\mu_0/\epsilon_0)} \approx 376.73 \Omega$;
- base mass scale $M_0 c^2 = \hbar c / l_0 \approx 429.5068 \text{ eV}$;
- Cosserat coupling $\mu_c = \eta = 2\pi$;
- elastic anisotropy $K_3/K_1 - 1 = \eta$, with $K_1 = K_2 = 2$, $K_3 = 2(1 + 2\pi) \approx 14.566$;
- Skyrme term $c_4 = 1$.

Minimization by the Nelder-Mead simplex method on a stretched cylindrical grid of resolution 1024×2048 gives:

$m_e^{\text{bare}} c^2 = 507.9966 \text{ keV}$

This value is interpreted as the **bare** (UV) electron mass in the Cosserat vacuum — in parallel with the bare $\alpha_{\text{bare}} = 1/128 = 2^{-7}$ from [1]. The corresponding physical (IR-measured) m_e (CODATA) = 510.999 keV differs by $\Delta = -3.002 \text{ keV}$, which matches $7 \cdot M_0 c^2$ to within 4 eV ($\sim 1.3 \cdot 10^{-3}$ of the gap, or $\sim 8 \cdot 10^{-6}$ of m_e).

Hypothesis (§7.2): this gap is the standard QED renormalization acting uniformly on both bare parameters ($\alpha_{\text{bare}} \rightarrow \alpha(0)$ and $m_e^{\text{bare}} \rightarrow m_e^{\text{phys}}$); a direct derivation of the δm_e contribution within the Cosserat program is an open direction.

Topology is preserved: $|Q_H| = 1$ to 10^{-5} ; the sign -1 corresponds to the electron.

Logical chain of the paper

```
INPUTS (three measured vacuum constants)
{ε₀, μ₀, ħ}                                [1, §6]
|
▼
DERIVED CONSTANTS (from [1])
c      = 1/√(ε₀μ₀)                          [1, §2.4]
Z₀     = √(μ₀/ε₀) ≈ 376.73 Ω                [1, §6]
l₀⁴    = ħ/(2π Z₀) → l₀ ≈ 4.5943 Å          [1, §7]
M₀ c²  = ħc/l₀ ≈ 429.5068 eV                [1, §5.1]
|
▼
COSSERAT-FUNCTIONAL PARAMETERS (§3)
η      = 2π (zero-point oscillations of spin waves)
μ_c    = η = 2π (Cosserat coupling)
K₃/K₁ - 1 = η (elastic anisotropy, K₁=K₂=2, K₃=14.566)
c₄     = 1 (Skyrme stabilizer)
|
▼
TOPOLOGICAL CONSTRAINT (§4)
Q_H = 1 (Hopf invariant, π₃(S²) = ℤ)
3-parameter Hopf ansatz n(r, z; R_r, R_z, w)
|
▼
NUMERICAL MINIMIZATION (§5)
Nelder-Mead on stretched grid 1024×2048
(R_r, R_z, w) = (0.51688, 0.76148, 0.62580) (in units of l₀)
|
▼
RESULT (§6)
```

```

m_e^bare c^2 = M_0 c^2 · Ê_min = 429.5068 eV · 1182.74 = 507.997 keV
– the bare electron mass in the Cosserat vacuum
Q_H = -0.99999 (electron orientation; |Q_H| = 1 preserved to 10-5)
|
▼
INTERPRETATION (S7)
m_e^bare = 507.997 keV – bare value
Gap to CODATA (510.999 keV) = 3.002 keV ≈ 7·M_0c^2
Hypothesis (S7.2): standard QED renormalization, acting in
parallel on α_bare → α(0) and m_e^bare → m_e^phys

```

Minimization is reproducible via the standalone script

verifications/electron_mass_minimization/nm_minimization.py (see Appendix A).

1. Introduction

1.1. The electron mass in the Standard Model

$m_e c^2 = 510.998950 \pm 0.000015$ keV enters the Standard Model as an empirical parameter. The Yukawa coupling $y_e \approx 2.94 \times 10^{-6}$ is fixed by this value; its smallness relative to the top-quark Yukawa ($y_t \approx 1$) has no independent explanation.

Earlier attempts to derive m_e from deeper structure (Eddington 1924 [9], Wyler 1969 [10], Atiyah 2018 [12]) were either numerological or relied on arbitrary normalization choices.

1.2. The Cosserat-vacuum hypothesis

The preceding work [1] showed that, under four structural identifications, the SI base reduces to a single dimension (energy), and the electromagnetic and gravitational sectors are accommodated in a single Cosserat continuum. The functional $E[n, u]$ in this medium admits localized minima with integer $Q_H \in \pi_3(S^2) = \mathbb{Z}$. The minimum $Q_H = -1$ is identified with the electron.

The present work computes the mass of this minimum.

1.3. Approach

1. The Cosserat functional is constructed; all its constants are fixed by the triple $\{\epsilon_0, \mu_0, \hbar\}$ (S3).
2. Minimization is performed by the Nelder-Mead method over the 3-param Hopf ansatz on a 1024×2048 grid (S§4–5).
3. The minimum is converted to SI units via $M_0 c^2 = \hbar c / l_0$ (S6).

The result is the bare electron mass $m_e^{\text{bare}} c^2 = 507.997$ keV. The gap from the measured m_e (CODATA) = 510.999 keV is 3.002 keV $\approx 7 \cdot M_0 c^2$ and is interpreted (S7.2) as the standard QED renormalization acting in parallel on $\alpha_{\text{bare}} = 2^{-7} \rightarrow \alpha(0) = 1/137$.

2. The Cosserat functional

A Cosserat medium is described by two independent field variables: a microrotation parameterized by the director $n(r) \in S^2$ and a translational displacement $u(r) \in \mathbb{R}^3$. The energy functional decomposes into three blocks:

$$E[n, u] = E_n[n] + E_u[u] + E_{\text{int}}[n, u] \quad (2.1)$$

Block E_n — energy of the director channel (Frank-Oseen + Skyrme):

$$\begin{aligned}
E_n[n] = \int d^3r [& \\
& (K_1/2) (\nabla \cdot n)^2 & \leftarrow \text{splay} & (\text{Frank, 1958}) \\
& + (K_2/2) (n \cdot \nabla \times n)^2 & \leftarrow \text{twist} & (\text{Frank, 1958}) \\
& + (K_3/2) (n \times \nabla \times n)^2 & \leftarrow \text{bend} & (\text{Frank, 1958}) \\
& + (c_4/4) [(\nabla n)^2 \otimes (\nabla n)^2] & \leftarrow \text{Skyrme stabilizer} & (\text{Skyrme, 1961}) \\
&] & & (2.2)
\end{aligned}$$

Block E_u — energy of the translational channel (elasticity + Yukawa mass term):

$$E_u[u] = \int d^3r [(\mu/2) (\partial u)^2 + (\mu_c/2) u^2] \quad (2.3)$$

Block E_int — linear $\mathbf{n} \rightarrow \mathbf{u}$ coupling (geometric source from director topology):

$$E_{\text{int}}[\mathbf{n}, \mathbf{u}] = - \int d^3r [2 \cdot f(\mathbf{n}) \cdot \mathbf{u}], \quad f(\mathbf{n}) = \text{atan2}(\sqrt{n_1^2 + n_2^2}, n_z) \quad (2.4)$$

Here $n_\infty = \hat{2}$ is the boundary value at infinity; $f(\mathbf{n})$ is the polar angle of the director from the vacuum direction (0 at infinity, π at the knot core); μ is the elastic modulus (in natural units $\mu = 1$).

3. The constants from $\{\epsilon_0, \mu_0, \hbar\}$

3.1. Length scale l_0

$$l_0^4 = \hbar / (2\pi \cdot Z_0), \quad Z_0 = \sqrt{(\mu_0/\epsilon_0)} \approx 376.730 \, \Omega \quad (3.1)$$

$$\Rightarrow l_0 \approx 4.5943 \, \text{\AA}$$

3.2. Base mass scale M_0

$$M_0 c^2 = \hbar c / l_0 = (2\pi \hbar^3 / (c^5 \epsilon_0))^{1/4} c^2 \quad (3.2)$$

$$M_0 c^2 \approx 429.5068 \, \text{eV}$$

3.3. Cosserat coupling μ_c

$$\mu_c = \eta = 2\pi \quad (3.3)$$

The dimensionless constant η is defined as the **ratio of the intrinsic quantum energy of a single cell of the medium to its elastic energy**:

$$\eta = \hbar \omega_{\text{gyro}} / (K \cdot l_0^3)$$

Each factor has a direct physical meaning.

Numerator: $\hbar \omega_{\text{gyro}}$ — the minimum-energy quantum of the gyroscope.

The frequency ω_{gyro} follows from the quantization of the angular momentum of a gyroscope of size l_0 : the moment of inertia $I \sim m_0 \cdot l_0^2$ and the angular momentum $L \sim \hbar$ give

$$\omega_{\text{gyro}} = L/I = \hbar / (m_0 \cdot l_0^2) = c/l_0$$

(using the identity $m_0 \cdot l_0 = \hbar/c$). The energy of one rotation quantum is

$$\hbar \omega_{\text{gyro}} = \hbar c / l_0 = M_0 c^2 \approx 429.5 \, \text{eV}$$

Denominator: $K \cdot l_0^3$ — the elastic energy of one cell.

K is the Frank constant with dimensions [energy/length]. It is related to the base shear modulus by $K = \mu \cdot l_0$, where $\mu \equiv 1/\epsilon_0$ (postulate P2 of [1]). Therefore $K \cdot l_0^3 = \mu \cdot l_0^4$ — the energy stored when the field $\mathbf{n}$ is deformed by $O(1)$ over the volume of one cell.

Substituting the fundamental length scale $l_0^4 = \epsilon_0 \hbar c / (2\pi)$ (see (3.1)) gives:

$$\begin{aligned} \eta &= \hbar c / (\mu \cdot l_0^4) \\ &= \hbar c \cdot \epsilon_0 \cdot (2\pi) / (\epsilon_0 \cdot \hbar c) \\ &= 2\pi \end{aligned}$$

This is not a free parameter but an algebraic consequence of the triple $\{\epsilon_0, \mu_0, \hbar\}$. The physical meaning of the result: in the vacuum Cosserat medium the quantum "trembling" of the field $\mathbf{n}$ and the elastic stiffness of the medium are of the same order and differ by exactly the factor 2π .

3.4. Elastic anisotropy K_3/K_1

$$K_1 = K_2 = 2, \quad K_3 = K_1 \cdot (1 + \eta) = 2 \cdot (1 + 2\pi) \approx 14.566 \quad (3.4)$$

3.5. Skyrme stabilizer c_4

$$c_4 = 1 \quad (3.5)$$

3.6. Summary

All five parameters of the functional $\{K_1, K_2, K_3, \mu_c, c_4\} = \{2, 2, 14.566, 2\pi, 1\}$ are fixed by $\{\varepsilon_0, \mu_0, \hbar\}$ via the identities (3.1)–(3.5).

4. The Hopf ansatz

4.1. Topological context

A smooth field $n: \mathbb{R}^3 \rightarrow S^2$ with $n(r \rightarrow \infty) = \hat{z}$ extends to $S^3 \rightarrow S^2$. Such maps are classified by $\pi_3(S^2) = \mathbb{Z}$; the integer label is the Hopf invariant Q_H . For the electron identification the relevant minimum is $Q_H = -1$.

4.2. Three-parameter ansatz

$$\begin{aligned} n_1 &= 8 r z w / P \\ n_2 &= -4 r Y w / P \\ n_3 &= (D - 4 r^2 w^2) / P \end{aligned} \quad (4.2)$$

where:

$$\begin{aligned} Y &= r^2 + z^2 - 1 && \text{in scaled coordinates } (r/R_r, z/R_z) \\ D &= 4 z^2 + Y^2 \\ P &= D + 4 r^2 w^2 \end{aligned}$$

Three parameters: R_r (ring radius), R_z (ring height), w (twist amplitude). The ansatz carries $|Q_H| = 1$ for any positive (R_r, R_z, w) and reduces to $n_3 = 1$ at infinity. The sign of Q_H is set by the orientation of the field; the identification with the electron uses $Q_H = -1$ (positron: $+1$). The numerical implementation (§5) yields $Q_H = -0.99999$ (§6.2) according to the convention used in the code.

5. Numerical method

5.1. Grid

A cylindrical grid (r_i, z_j) of size $N_r \times N_z = 1024 \times 2048$ with clustering near $r \sim R_r$ and $z \sim 0$ (stretching exponents $\beta_r = 6.0$, $\beta_z = 3.0$). Box dimensions: $L_r = 24 l_0$, $L_z = 48 l_0$. Double precision (float64) throughout.

5.2. Optimizer

Minimization is performed using `scipy.optimize.minimize` with `method='Nelder-Mead'`, `xatol = 1e-5`, `fatol = 1e-3` (keV). The choice of method is motivated by the following considerations:

1. The energy landscape over (R_r, R_z, w) is smooth but non-convex.
2. The topological constraint $Q_H = 1$ is enforced exactly by the ansatz and requires no separate monitoring.
3. Gradient-based methods over the full field (Adam, L-BFGS) on a discrete grid induce drift of Q_H away from its integer value.

5.3. Convergence

$$\begin{aligned} R_r &= 0.51688 \quad (\text{in units of } l_0) \\ R_z &= 0.76148 \\ w &= 0.62580 \end{aligned} \quad (5.1)$$

5.4. Boundary conditions in z

For the z -derivatives, ghost cells constructed via linear extrapolation are used:

$$\text{padded}[Nz+i] = (i+1) \cdot \text{tensor}[Nz-1] - i \cdot \text{tensor}[Nz-2]$$

An alternative choice — periodic BC (via `torch.roll`) — is incorrect for this problem: the Hopf field is not periodic in z , and the periodic wrap introduces a jump at the boundaries $z = \pm L_z/2$. This jump contributes to the bend term and the contribution grows with grid refinement, leading to a divergence of the result under resolution increase. With linear extrapolation the grid convergence is preserved (< 1 eV per doubling).

5.5. Computing E_u : elimination of the u-channel

Numerically we solve the problem honestly: for each n the optimal u is found from the Euler-Lagrange equation of the functional (2.1)–(2.4) and substituted back into the energy.

Derivation of the equation. The u -dependent part of the functional is $E_u[u] + E_{\text{int}}[n, u]$:

$$\int d^3r \left[(\mu/2)(\partial u)^2 + (\mu_c/2) u^2 - 2 \cdot f(n) \cdot u \right] \quad (5.2)$$

with $\mu = 1$ in natural units. The minimization condition $\delta/\delta u = 0$ gives the linear equation

$$(\nabla^2 - \mu_c) u = -2 \cdot f(n) \quad (5.3)$$

— a screened Poisson equation with screening length $\lambda_c = 1/\sqrt{\mu_c} \approx 0.399 \lambda_0$. The term $-\mu_c$ plays the role of a **Yukawa mass for the u-channel**: the displacement u is exponentially suppressed on the scale λ_c outside the region where n deviates from n_∞ . The source $-2 \cdot f(n)$ is geometric and μ_c -independent — it reflects the pure topology of the knot.

Axisymmetric representation. In cylindrical geometry one introduces a stream function $\psi(r, z)$ that automatically enforces $\nabla \cdot u = 0$:

$$u_r = -\partial\psi/\partial z, \quad u_z = \psi/r + \partial\psi/\partial r \quad (5.4)$$

Substituting (5.4) into (5.3) yields a scalar equation for ψ :

$$(\partial_r^2 + (1/r) \partial_r - 1/r^2 + \partial_z^2 - \mu_c) \psi = -2 f \quad (5.5)$$

The $-1/r^2$ term arises from the curvature of cylindrical basis vectors when the vector Laplacian acts on the azimuthal component.

Numerical solution. Equation (5.5) is solved by preconditioned conjugate gradients (PCG, diagonal preconditioning) on a uniform subgrid 512×1024 with extents $L_r^{\{u\}} = 6 \lambda_0$, $L_z^{\{u\}} = 12 \lambda_0$. The compactness of the subgrid is justified by the exponential decay of the solution on the scale $\lambda_c \ll L_r^{\{u\}}$. The source f is bicubically interpolated from the stretched n -grid onto the uniform ψ -grid. The PCG relative residual is driven down to 10^{-6} .

Energy assembly. From the resulting ψ the components (u_r, u_z) are reconstructed via (5.4), followed by the strain tensor in axisymmetry:

$$\begin{aligned} \varepsilon_{rr} &= \partial u_r / \partial r, & \varepsilon_{zz} &= \partial u_z / \partial z, & \varepsilon_{\phi\phi} &= u_r / r, \\ \varepsilon_{rz} &= \frac{1}{2} (\partial u_r / \partial z + \partial u_z / \partial r) \end{aligned} \quad (5.6)$$

and the elastic energy density:

$$E_u = \int (\varepsilon_{rr}^2 + \varepsilon_{zz}^2 + \varepsilon_{\phi\phi}^2 + 2 \varepsilon_{rz}^2) \cdot 2\pi r \, dr \, dz \quad (5.7)$$

This is the quantity $E_u = 48.85$ keV of the table in §6.3 — the gravitational self-energy of the u-channel, induced by the presence of the EM-topology n .

Remark on limits. In the limit $\mu_c \rightarrow 0$ ($\lambda_c \rightarrow \infty$), equation (5.3) loses its screening while the source $-2f$ is retained, and E_u diverges in the UV like the self-energy of a point charge in classical electrodynamics. In the opposite limit $\mu_c \rightarrow \infty$ ($\lambda_c \rightarrow 0$), the mass term pushes u toward zero faster than the source can excite it: $u \approx 2f/\mu_c \rightarrow 0$, and $E_u \rightarrow 0$. The canonical value $\mu_c = 2\pi$ (3.3) provides $\lambda_c \approx R_r$ — the unique regime in which the u-channel contributes a finite nontrivial $\sim 10\%$ of the mass. Physically this means that **the $n \rightarrow u$ coupling cannot be switched off by setting μ_c to zero** (this, on the contrary, drives E_u to infinity): the source from the topology of n is always present, and μ_c only regulates the degree of its localization.

6. Results

6.1. Predicted electron mass

The functional minimum gives:

$$\begin{aligned}\tilde{E}_{\min} &= 1182.74 \quad (\text{dimensionless}) \\ E_{\min} &= M_0 c^2 \cdot \tilde{E}_{\min} = 429.5068 \text{ eV} \cdot 1182.74 = 507.9966 \text{ keV}\end{aligned}\quad (6.1)$$

Compared with CODATA 2018:

$$\begin{aligned}\text{Prediction:} \quad m_e c^2 &= 507.9966 \text{ keV} \\ \text{CODATA 2018:} \quad m_e c^2 &= 510.998950 \pm 0.000015 \text{ keV} \\ \text{Deviation:} \quad \Delta &= -3.002 \text{ keV} = -0.587 \% \text{ from CODATA}\end{aligned}\quad (6.2)$$

6.2. Topological charge preservation

$$Q_H = -0.99999 \quad (\text{electron, } |Q_H| = 1 \text{ to } 10^{-5}) \quad (6.3)$$

6.3. Energy decomposition

E_{OF}	(Frank–Oseen, $K_1 + K_2 + K_3$)	:	180.985 keV	(35.6 %)
E_{Sk}	(Skyrme stabilizer, c_4)	:	276.932 keV	(54.5 %)
E_u	(Cosserat coupling, μ_c)	:	50.080 keV	(9.9 %)
Total		:	507.997 keV	(100 %) (6.4)

The Skyrme term carries 54.5 % of the energy — the standard Derrick balance: under uniform rescaling Frank–Oseen scales as λ and Skyrme as λ^{-1} , and both are required at comparable magnitude to fix the soliton size [15]. The Cosserat coupling E_u contributes 9.9 %.

6.4. Grid convergence

Doubling the grid $1024 \rightarrow 2048$ shifts m_e by less than 1 eV — four orders of magnitude smaller than the gap to CODATA itself (3.002 keV). Grid convergence is stable, as is the conservation of the topological charge ($|Q_H| = 1$ to 10^{-5}).

6.5. Sensitivity analysis

From the identities of §3 the predicted scaling is:

$$m_e c^2 \propto \hbar^{(3/4)} \cdot \epsilon_0^{(-5/8)} \cdot \mu_0^{(-3/8)} \cdot \tilde{E}_{\min} \quad (6.5)$$

The dimensionless minimum $\tilde{E}_{\min}$ depends only on the dimensionless parameters (K_i, μ_c, c_4) and is invariant under rescaling of $\{\epsilon_0, \mu_0, \hbar\}$. Sensitivity to +1 % perturbations of each input constant:

$$\begin{aligned}\delta(\epsilon_0)/\epsilon_0 = +1 \% &\rightarrow \delta(m_e c^2) / m_e c^2 \approx -0.625 \% \\ \delta(\mu_0)/\mu_0 = +1 \% &\rightarrow \delta(m_e c^2) / m_e c^2 \approx -0.375 \% \\ \delta(\hbar)/\hbar = +1 \% &\rightarrow \delta(m_e c^2) / m_e c^2 \approx +0.750 \%\end{aligned}\quad (6.6)$$

7. Discussion

7.1. The 3-keV gap from CODATA

The gap $\Delta = -3.002 \text{ keV}$ (−0.587 %) is the difference between the **bare** mass $m_e^{\text{bare}} = 507.997 \text{ keV}$ obtained by minimizing the Cosserat functional and the physical $m_e^{\text{phys}} = 510.999 \text{ keV}$ measured experimentally. In §7.2 we put forward the hypothesis that this difference is the standard QED renormalization acting in parallel on both bare parameters ($\alpha_{\text{bare}} = 1/128 \rightarrow \alpha(0) = 1/137$ and $m_e^{\text{bare}} \rightarrow m_e^{\text{phys}}$). A direct derivation of the δm_e contribution within the Cosserat program is an open direction for future work.

7.2. Numerical observation: gap ≈ 7 quanta of $M_0 c^2$

In units of the base energy quantum $M_0 c^2 = 429.5068 \text{ eV}$:

$$\Delta / m_e c^2 = 3002.4 / 429.5068 = 6.99$$

In other words, the gap matches 7 quanta of $m_e c^2$ to within ~ 4 eV (~ 0.001 % of m_e).

A geometric check: the volume of the region in which the director is tilted by more than 60° from the vertical ($n_z < 0.5$) at the canonical minimum is $6.96 \cdot 10^{-3} \approx 7$ elementary cells of the medium — the same "7" as in the energy gap.

Below we put forward a hypothesis on the nature of this coincidence via the standard QED renormalization; a direct derivation of the δm_e contribution from the Cosserat functional is not performed in the present work and is left for future work.

For context: in [1] the **bare** value of the fine-structure constant is shown to be $\alpha_{\text{bare}} = 1/128 = 2^{-7}$. This value is systematically lower than the measured $\alpha(0) = 1/137$ by ≈ 6.6 %, which in standard QED is explained by the running coupling (vacuum polarization by virtual e^+e^- pairs). Our $m_e = 507.997$ keV is in the same situation: it is systematically lower than the measured m_e (CODATA) = 510.999 keV by 0.587 %. **Hypothesis:** both values are bare (UV) and the discrepancies with experiment reflect the same QED renormalization acting on both parameters. In this case $7 \cdot m_e c^2$ may be a numerical manifestation of the mass-renormalization contribution $\delta m_e \sim (\alpha/\pi) \cdot m_e \cdot 0(1)$ with the cutoff at the scale $\Lambda \sim 1/\ell_0$. A direct calculation of this contribution within the Cosserat program is an open direction for future work.

Implication for the interpretation of precision. If this hypothesis holds, the formal "0.587 % from CODATA" reflects **not** the accuracy of the calculation, but the **structurally expected** QED correction distinguishing the bare m_e^{bare} from the physical m_e^{phys} . The match $\Delta \approx 7 \cdot m_e c^2$ holds to within ≈ 4 eV — $\sim 1.3 \cdot 10^{-3}$ of the gap itself, or $\sim 8 \cdot 10^{-6}$ of m_e . In other words, our bare value 507.997 keV is accurate at the level of grid convergence (~ 1 eV, see §6.4), while the 3.002 keV gap is a separate, structurally determined object, **not a residual error**.

7.3. Position within the Cosserat program

The present work belongs to the tradition of deriving fundamental physics from the topology of a continuous medium. The main representatives of this line are the program of **G. E. Volovik** on the analogy between superfluid $^3\text{He-A}$ and the Standard Model [19] and the program of **H. Kleinert** on multivalued fields, describing defects as gauge fields and gravity through torsion (Einstein-Cartan) [20]. The mathematical apparatus of all three programs is essentially common: Frank-Oseen-type elasticity for a director field on the sphere (for $^3\text{He-A}$ — the l-vector of the orbital part), topological invariants of homotopy classes of mappings, and hopfions as stable configurations.

The principal difference lies in the choice of substrate. Volovik works with a concrete quantum medium (superfluid $^3\text{He-A}$ at millikelvin temperatures), Kleinert — with an elastic continuum carrying dislocations and disclinations. The present work takes as the medium the physical vacuum itself: four identifications of medium parameters with the fundamental constants $\{\epsilon_0, \mu_0, \hbar, G\}$ (see [1]) turn the abstract continuum into a concrete model from which a numerical electron mass $m_e c^2 = 507.997$ keV ($\Delta = -0.587$ % from CODATA) is derived with zero free parameters.

More broadly, these programs share a related mathematical apparatus (Frank-Oseen elasticity for an S^2 -target field, topological invariants, hopfions). **One possible hypothetical unifying picture** is the KTHNY cascade of topological melting (Kosterlitz-Thouless-Halperin-Nelson-Young [22, 23]): a rigid crystal (Kleinert's "World Crystal"), an intermediate hexatic phase (liquid-crystalline, Frank-Oseen), and an isotropic liquid (close to superfluid $^3\text{He-A}$). Under this hypothetical picture, the present work would lie in the intermediate phase with particles as topological defects. **A rigorous verification of this picture for the physical vacuum** (including the numerical correspondence of the functional $E[n, u]$ to the relevant KTHNY phase) **is not performed here and is left as an open direction**; it is mentioned only as one possible broad context, not as a claim of the program.

8. Conclusion

The **bare** electron mass $m_e^{\text{bare}} c^2 = 507.9966$ keV is obtained numerically from the triple of vacuum constants $\{\epsilon_0, \mu_0, \hbar\}$ by minimization of the Cosserat functional over the 3-parameter Hopf ansatz. The result is stable under grid refinement (< 1 eV per doubling of the resolution) and invariant under grid/code rescaling (see §6.4).

The difference from the physical m_e (CODATA) = 510.999 keV is -3.002 keV and matches $7 \cdot m_e c^2$ to within 4 eV ($\sim 1.3 \cdot 10^{-3}$ of the gap). By parallel with the bare $\alpha_{\text{bare}} = 1/128 = 2^{-7}$ from [1], which differs from the measured $\alpha(0) = 1/137$ by ≈ 6.6 % (standard QED renormalization), **we put forward the hypothesis** (§7.2) that the gap $m_e^{\text{phys}} - m_e^{\text{bare}}$ is the same QED renormalization acting on both bare parameters. In this case the accuracy of the present calculation for the bare m_e corresponds to grid convergence (~ 1 eV, or $\sim 2 \cdot 10^{-6}$ of

m_e), and the formal " -0.587% " is a structurally expected correction, not a residual error.

A direct derivation of the δm_e contribution within the Cosserat program is an open direction for future work.

Acknowledgements

The conceptual basis of this work draws on the ideas of **L. D. Faddeev** on topological solitons as particles, the fourth-derivative stabilizing term introduced by **T. H. R. Skyrme**, and the elastic theory of director media due to **C. W. Oseen** and **F. C. Frank**.

The computations were performed on a single NVIDIA RTX 2070 using PyTorch.

The discretization bug in the z-boundary conditions was identified through comparison of results across multiple grid resolutions during a diagnostic session on 2026-05-16.

References

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-

Appendix A. Reproducibility recipe

```
verifications/electron_mass_minimization/
├── nm_minimization.py      # main minimization script
├── stretched_grid.py       # grid utilities, energy, Cosserat solver
├── requirements.txt
├── README.md
└── result.json             # output (created upon run)
```

Reproduction in three commands:

```
cd verifications/electron_mass_minimization
pip install -r requirements.txt
python nm_minimization.py
```

The script builds the Cosserat functional and minimizes by Nelder–Mead on the 1024×2048 grid from the standard initial simplex $(0.50, 0.70, 0.60)$.

Expected canonical output:

```
R_r ≈ 0.51688,    R_z ≈ 0.76148,    w ≈ 0.62580
Q_H ≈ -0.99999
E_OF = 180.985 keV (35.6 %)    E_Sk = 276.932 keV (54.5 %)
E_u = 50.080 keV ( 9.9 %)
E_total = 507.997 keV (= m_e^bare; CODATA m_e = 510.999 keV, see §7.2)
```

A typical run converges in ~ 120 simplex iterations; wall time on a single NVIDIA RTX 2070 is of order 3–5 minutes.

Appendix B. The functional in dimensionless form

In dimensionless variables ($\tilde{r} = r/l_0$) the functional of §2 takes the form:

$$\tilde{E}[n, u] = \tilde{E}_n[n] + \tilde{E}_u[u] + \tilde{E}_{\text{int}}[n, u] \quad (\text{B.1})$$

$$\begin{aligned} \tilde{E}_n[n] = \int d^3\tilde{r} [& \\ & (K_1/2) (\nabla \cdot \tilde{n})^2 \\ & + (K_2/2) (\tilde{n} \cdot \nabla \tilde{n})^2 \\ & + (K_3/2) (\tilde{n} \times \nabla \tilde{n})^2 \\ & + (c_4^2/4) [(\nabla \tilde{n})^2 \otimes (\nabla \tilde{n})^2] \\ &] \end{aligned} \quad (\text{B.2})$$

$$\tilde{E}_u[u] = \int d^3\tilde{r} [(1/2) (\partial \tilde{u})^2 + (\tilde{\mu}_c/2) u^2] \quad (\text{B.3})$$

$$\tilde{E}_{\text{int}}[n, u] = - \int d^3\tilde{r} [2 \cdot f(\tilde{n}) \cdot u] \quad (\text{B.4})$$

with dimensionless coefficients $\{K_1, K_2, K_3, \tilde{\mu}_c, c_4^2\} = \{2, 2, 14.566, 2\pi, 1\}$ (§3).

The mass in physical units: $m_e^{\text{bare}} c^2 = M_0 c^2 \cdot \tilde{E}_{\text{min}}$.

Appendix C. Data and code availability

The code is in the repository:

<https://github.com/igorevsiev-cmyk/cosserat-program>

Supplementary materials in [papers/2026-05-electron-mass/](#). The preprint is accompanied by numerical checks:

- [verifications/electron_mass_minimization/](#) — the Nelder–Mead minimization (reproduces $m_e = 507.997$ keV);
- [verifications/canonical_derrick/](#) — Derrick stability scan.

The preprint is registered on Zenodo: [10.5281/zenodo.20205502](<https://doi.org/10.5281/zenodo.20205502>).

The preceding work [1] is registered with DOI: [10.5281/zenodo.20187199] (<https://doi.org/10.5281/zenodo.20187199>).

